Bryan Larsen

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(613) 709-1523, bryan@larsen.st

Full stack: from circuit design to CSS. From frame buffer to cloud.

Career Highlights

- 4 years of production Rust: designed and built a cross-platform zero-trust network security agent, shipped on Windows, macOS and Linux
- Wrote all the firmware for version 1.0 of Lumenera's Ethernet IP video cameras: embedded Linux on ARM, streaming the SOC's JPEG captures over the network as RTSP or HTTP motion-JPEG
- Designed the collaborative workspaces, job system and caches behind Clara.io, a browser-based 3D modelling and cloud-rendering platform
- Very early employee or cofounder at 5 startups
- 30+ years of Unix and programming experience
- Early git contributor; made git/Cogito work on Mac OS X and maintained MacPorts port
- Video game in assembly language sold to distributor in middle school

Expertise

10+ years in each

- **Systems & Networking:** Rust, Tokio, WireGuard, DNS, IPv6, routing, netlink, FIPS 140-3, high-availability job systems, multi-level caching
- **Full Stack:** React, Typescript, Javascript, CSS, Node.js, Ruby, Rails, Axum, Tauri, Webpack, Selenium, CGI, SQL, PostgreSQL, MySQL, MongoDB, Redis, SQLite
- **DevOps:** Kubernetes, Docker, NixOS, GCP, AWS, Prometheus, Loki, Grafana, Ansible, Jenkins, Make, sh, Bash, TCL, Git, GitLab CI/CD
- **Embedded:** C, C++, Linux Drivers, U-boot, ARM/68K/8051 assembly, RTSP, HTTP, Circuit Design, VHDL

Links

- Stack Overflow
- GitHub
- LinkedIn

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Bowtie 2022-2026

Senior Software Engineer Remote
Network Security for Aerospace

- Designed and developed the cross-platform zero-trust VPN client in Rust, a WireGuard-based agent with split DNS and policy enforcement
- Developed the FIPS 140-3 compliance implementation
- Signed Windows, MacOS and Linux deb/rpm installation packages
- Code review support on the entirety of the Bowtie platform
- Rust, Python, Javascript, DNS, WireGuard, Automerge CRDT, Tauri, Tokio, Axum, SQLite, MSI, IPv6, NixOS, netlink, routing, FIPS, signing, Prometheus, Loki, Grafana

Threekit / Clara.io 2012-2022

Senior Software and DevOps Engineer Ottawa
Online 3D Modelling Software

- Designed the collaborative workspaces, job system and caches
- Led DevOps team
- Javascript, Typescript, ES6, Node.js, WebGL, V-Ray, Operational Transforms, HTML, CSS, Backbone.js, React, Mongo, Redis, Prometheus, nginx, Ansible, Docker, Jenkins, beanstalkd, statsd, AWS, Vagrant, Kubernetes

Barquin 2009-2013

contractor
NIFA web applications

- Ruby, Rails, Javascript, jQuery, Selenium, SQL, MySQL

Hobo, hobo-jquery 2009-2013

maintainer
popular open source web framework

- core developer and release manager
- Ruby, Rails, Javascript, prototype.js, jQuery, selenium, SQL, MySQL, SQLite, git

Magor Corporation 2008

contractor Kanata, Ontario
Phone based control panel software

- Javascript, Groovy, Java, HTML
- JETTY, SIP, Mitel

Lumenera Corporation 2002 - 2006

Senior Software Engineer Ottawa, Ontario
Ethernet IP Video Cameras

- Wrote all the firmware for camera 1.0, streaming the SOC's JPEG captures as RTSP or HTTP motion-JPEG
- Assisted with hardware design
- Designed and deployed initial software design process and infrastructure
- Ported Linux and U-boot to underdocumented ARM SOC
- Power on Self-Test design and implementation

- Manufacturing System Test and Customization design and implementation
- Worked with manufacturer to ensure reliable production
- Early employee: wide variety of business experiences, including hiring, firing, customer interaction, financial modelling, assembling
- C, Javascript, HTML, sh, Python, ARM & 68K assembler, C++, make
- JPEG, motion-JPEG, MPEG, EXIF, TIFF
- Linux Kernel and Drivers, U-Boot, CGI, XML-RPC, AJAX, Boa (web server), RTP/RTSP, NTP, zeroconf, ActiveX, HTML, XHTML, CSS, Twisted
- Bugzilla, Subversion, TWiki, PyUnit, CUnit, MySQL, Apache

Analog Design Automation (currently Synopsys) 1999 - 2002

Senior Software Engineer Saskatoon, Saskatchewan and Ottawa, Ontario

Analog Integrated Circuit Synthesis and Optimization Software

- Responsible for initial design and implementation of Parts Library
- Responsible for integration with Cadence Analog Artist
- Led small team in an Extreme Programming development of a Synthesis Prototype
- C++, Python, TCL, CORBA, LISP
- Cadence Analog Artist, SPICE

Vitana Corporation (DBA Pixelink) 1998 - 1999

Senior Software Engineer Ottawa, Ontario

IEEE-1394

- Solely responsible for design and implementation of an embedded IEEE-1394 stack
- Project lead and customer interface for design of airline set top box
- C, 8051 assembler

Newbridge Networks (currently Alcatel-Lucent) 1997 - 1998

Junior Hardware Engineer Ottawa, Ontario

Time Division Multiplexing

- Responsible for implementation of Data Multiplex FPGA and Glue Logic FPGA
- VHDL, Schematic Capture
- Mentor Graphics

Co-op Jobs for Kipp & Zonen, Ricoh, SED Systems, Environment Canada 1993 - 1997

- 8051 assembler, C, C++, FORTRAN, Visual BASIC
- Labview, COM, OLE, DLL, MFC, Watcom 4G
- HP-UX, OS/2, MS-DOS, Data General AOS/VS

T&D Software 1987 - 1990

Freelance Designer Holland, MI

Video Games

- Design and Implementation of Video Games
- 6809 Assembler, BASIC

Education

- University of Regina, 1997, Bachelor of Applied Science in Electronic Systems Engineering, Honours